

# **Design of a Vending Machine using Finite Automata**

Theory of Computing (TOC) Project Report

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## **Abstract**

This report presents a detailed study on modeling a vending machine using finite automata. The system is analyzed using DFA, transitions, and formal definitions. The goal is to demonstrate how theoretical models can simulate real-world systems effectively.

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## Introduction

Finite automata are used to represent computational systems with limited states. A vending machine is an example where inputs produce outputs based on state transitions. This section explains the background, importance, and objectives of the study.

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## **Problem Statement**

The vending machine accepts coins and dispenses items when required conditions are met. This section defines constraints, assumptions, and expected behavior of the system.

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## Theoretical Background

A deterministic finite automaton (DFA) is defined by a 5-tuple. It consists of states, transitions, input alphabet, start state, and final states. This section explains all concepts in detail.

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## Methodology

The system is modeled step by step by defining states, transitions, and acceptance criteria. State diagrams and tables are constructed for clarity.

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## Model Design

The vending machine is represented using states q0, q1, q2, and q3. Transitions occur based on coin inputs. Final state triggers product dispensing.

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## Implementation

The model can be simulated using programming tools. This section explains logic, pseudo-code, and simulation approach.

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## **Analysis and Discussion**

The DFA model is analyzed for correctness and efficiency. Different input cases are evaluated and validated.

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## Applications

Finite automata are widely used in real-world systems like vending machines, compilers, and ATMs.

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## **Conclusion**

The project successfully demonstrates DFA application in vending machines.

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## **Future Scope**

The model can be extended to more complex automata and real-world applications.

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## References

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